

Learning Streaming Video Representation via Multitask Training

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Background: Streaming Video Understanding is Transforming the World

- We target real-time perception for robotics, autonomous driving, AR/VR, and human–AI interaction.
- We must make decisions frame by frame under strict latency and memory constraints.
- We seek to maintain long-term temporal context while preserving fine spatial detail.

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The Problem: Offline Video Models Don't Fit Streaming Constraints

- We find most video backbones use bi-directional attention over clips—impractical for causal, per-frame inference.
- We observe that aggregating frames with image encoders loses critical temporal and spatial precision.
- We need a single backbone that is causal, efficient, and preserves strong image features.

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Our core idea: StreamFormer + multitask visual–language alignment

- We inject causal temporal attention into a strong ViT to enable low-latency streaming.
- We train a single backbone across global, temporal, and spatial tasks via video–text alignment.
- We freeze and transfer to diverse real-time tasks: OAD, online VIS, and VideoQA.

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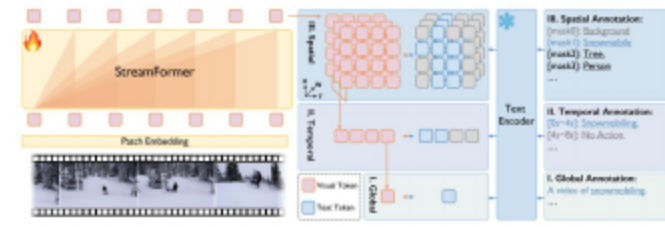
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- We build StreamFormer: causal temporal attention + spatial LoRA on a SigLIP ViT.
- We unify multitask video-text alignment across global, temporal, and spatial granularities.
- We train data-efficiently with curated annotations, not web-scale video-text.
- We match or surpass SOTA with frozen-backbone transfer; we keep near-constant latency via a KV-cache.

Backbone design: causal temporal attention for streaming

- We use divided space-time Transformer blocks with a strictly causal mask over time.
- We let each frame attend only to historical keys/values; future frames are excluded.
- We cache keys/values (KV-cache) to enable near-constant per-frame latency for long horizons.

- We tokenize frames with spatial/temporal embeddings; we apply causal temporal then spatial attention.
- We produce three feature granularities: global (last frame), temporal (per frame), spatial (per patch).
- We align all features with a shared text space for multi-task training.



Preserving image strength: we initialize from SigLIP and adapt with LoRA

- We initialize spatial layers from a SigLIP ViT to retain strong image representations.
- We adapt efficiently with spatial LoRA (rank 32) for video-specific tuning.
- We add a learnable tanh gate (init=0) to gradually introduce temporal interactions during training.

Multi-granularity features for many tasks

- Global feature v (last frame) for causal global semantics (e.g., AR, VTR).
- Temporal features f (per frame) for dynamics (e.g., TAL, TVG).
- Spatial features F (per patch) for fine-grained localization (e.g., VOS, RVOS).
- Qualitative: we track actions frame-by-frame without future context (e.g., action onset is correctly localized over the highlighted frames).
- Qualitative: for online VIS, we maintain instance masks through occlusions in streaming mode.



Unified visual-language alignment: global level

- We encode text prompts (e.g., "a video of kicking a football") via the SigLIP text encoder.
- We align last-frame video feature v with text embeddings for global tasks (AR/VTR).
- We use sigmoid-based alignment to preserve causality while learning semantics.

Unified visual-language alignment: temporal level

- We align per-frame features f to temporal text prompts/queries (TAL/TVG).
- We apply frame-wise sigmoid/contrastive objectives to capture event dynamics.
- We support fine temporal localization under streaming constraints.

Unified visual-language alignment: spatial level

- We align per-patch features F with text for pixel-wise logits (closed/open set).
- VOS: category text; RVOS: referring expressions for open-vocabulary segmentation.
- We upsample logits and train with cross-entropy/contrastive formulations.



- We alternate batches across task families and use gradient accumulation to reduce bias.
- We use balanced sampling and oversampling to handle dataset imbalances.
- We train a base SigLIP-224 for 20 epochs; we sample 16 frames per clip; batch size 128, LR $1e-6$.

- We evaluate Online Action Detection on THUMOS-14 (per-frame mAP), with/without optical flow.
- We evaluate Online Video Instance Segmentation on YouTubeVIS-2019 (AP, AR1) at 224p.
- We evaluate VideoQA on VideoMME and MLVU using LLaVA-Next (Vicuna-7B, 224×224).

Results in one slide: we improve accuracy and keep latency flat

- Online Action Detection (THUMOS-14): we raise mAP by +9.5 over SigLIP in the same head (68.4 vs 58.9 w/o flow); with flow, we reach 73.9.
- Online VIS (YouTubeVIS-2019, 224p): we improve AP to 45.1 (+4.2 over SigLIP) while keeping the backbone frozen.
- VideoQA (224×224): we outperform comparable 224p backbones on VideoMME/MLVU.
- Efficiency: with causal attention + KV-cache, we show near-constant per-frame latency; bi-directional attention grows quadratically.

Limitations and practical considerations

- Our current resolution (224×224) may limit fine-detail segmentation in some settings.
- Our instruction-following currently relies on the paired LLM; tighter coupling is possible.
- Our edge deployment can benefit from further hardware-aware optimization.

Conclusion: a single streaming backbone for many tasks

- We present StreamFormer: a causal, KV-cache-friendly ViT with preserved image strength.
- We unify multitask video-text alignment to learn global, temporal, and spatial cues.
- We transfer competitively to OAD, online VIS, and VideoQA with a frozen backbone.



Questions & discussion

- Thank you for your attention!
- We welcome questions and feedback.

Future work & impact

- We plan to scale to higher resolutions and longer horizons; we will integrate tracking and forecasting.
- We will tighten LLM coupling for instruction-following and grounding.
- We will optimize for hardware-aware deployment on edge/robotic platforms.

Acknowledgments

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Backup: key result—online action detection (Table 1)

- StreamFormer within MAT achieves 68.4 mAP (w/o flow) vs. SigLIP 58.9 (+9.5).
- With flow, we achieve 73.9 mAP—state-of-the-art among comparable settings.

Table: Table 1. Comparison on Online Action Detection.

Method	Backbone	THUMOS-14	
		w/ Flow	w/o Flow
TRN [109]	ResNet50	62.1	-
OADTR [42]	ResNet50	65.2	-
CoLAR [114]	ResNet50	66.9	-
LSTR [110]	ResNet50	69.5	63.5
GateHUB [16]	ResNet50	70.7	66.5
TesTra [129]	ResNet50	71.2	-
TRN [109]	TimeSformer	68.5	-

Backup: streaming efficiency and scalability

- Causal attention + KV-cache yields flat latency and memory as frames increase.
- Bi-directional attention grows quadratically—unsuitable for long-horizon streaming.

Table 5. Ablation study for temporal attention.

Figure: Computational complexity.

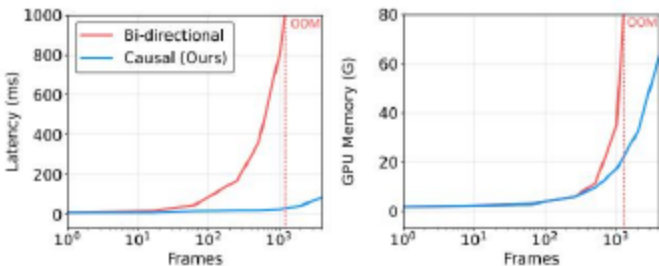


Figure: Computational complexity.

Backup: online VIS and VideoQA

- Online VIS (CTVIS, 224p): StreamFormer AP 45.1, AR1 43.6; surpasses SigLIP ViT and R50 under the same setup.
- VideoQA (LLaVA-Next, 224×224): 45.0 (VideoMME), 48.4 (MLVU)—best among 224p methods; matches higher-res baselines on MLVU.

Backup: ablation—pretraining task contributions

- Each granularity (global, temporal, spatial) alone improves over the SigLIP baseline.
- Global + temporal yields the strongest consistent gains for OAD and VideoQA.
- Adding spatial brings large VIS boosts while keeping strong OAD/VideoQA.

Table: Table 4. Ablation study for pre-training tasks.

Method	Global	Temporal	Spatial	Online Action Detection THUMOS-14		Online VIS		VideoQA	
				w/ Flow	w/o Flow	AP (YTVIS19)	AR1	VideoMME	MLVU
SigLIP				71.1	58.9	40.9	41.8	40.8	46.1
StreamFormer	!			73.1	65.9	41.9	41.6	43.7	44.6
StreamFormer		!		72.1	65.5	40.1	41.1	44.0	49.6
StreamFormer			!	70.3	51.5	42.3	41.6	30.0	29.2
StreamFormer	!	!		74.1	69.1	41.9	41.4	43.1	49.2
StreamFormer	!	!	!	73.9	68.4	45.1	43.6	45.0	48.4

- We find both bi-directional and causal temporal attention beat no temporal attention on OAD.
- We prefer causal attention for streaming due to KV-cache scalability.
- We accept small accuracy trade-offs for massive latency benefits online.

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